THE OPEN SOURCE AUDIT

Python

A Capstone Project for OSS NGMC Course

Course: Open Source Software Unit Coverage: 1 – 5

Max Marks: 100

Student Name	Registration Number
Anushka	24BAI10805
Slot	**Date of Submission**
D11	31 March 2026

Table of Contents

1. Introduction 3

2. Part A – Origin and Philosophy 3-6

2.1 Origin Story 3-4

2.2 License 4- 5

2.3 Ethics 5-6

Part B – Linux Footprint 6-8

Part C – Ecosystem 8-9

Part D – Comparison 9-11

Conclusion 11-12

3. Shell Script 12-16

3.1 Script 1 — System Identity Report 12-13

3.2 Script 2 — FOSS Package Inspector 13

3.3 Script 3 — Disk and Permission Auditor 14

3.4 Script 4 — Log File Analyzer 14-15

3.5 Script 5 — The Open Source Manifesto Generator 15-16

4. References Resources 16

`1. Introduction

Open source software is basically code that anyone can see,use,change and share.The source code is available publicly and people are free to download it,modity it,fix bugs and add features or even build entrire business around it.This is super different from closed source software where the company keeps the code locked away and you can only use it the way they allow.

What makes open source so cool is the collaboration and transparency it brings.Developers from all over the world,students,professionals work together on the same project.They share ideas,review each other's code,report issues and help make the software better over time.This leads to faster innovation,more reliable code and software that's free for everyone to use.

For beginners like me open source is a game changer for learning programming.You don't just read theory instead you can look at real code,see how professionals solve problems,experiments with changes and even contribute small fixes or features.It makes learning less scary.

In this project I picked **Python** as the open source software to study because it is one of the best examples of how powerful and beginner friendly open source can be.

2. Part A — Origin and Philosophy

A.1 Origin Story of Python

Back when programming was just starting to get popular most languages like C and C++ were really powerful but also super complicated for beginners. Writing even a simple program took a lot of effort the syntax was messy with all those semicolons and curly

braces and it was hard to read someone else's code. A lot of people (including me at first) found it confusing and took forever to learn.

Guido van Rossum a Dutch programmer working at a research institute felt the same way. He was frustrated with how hard it was to write clean, understandable code especially when he just wanted to focus on solving problems instead of fighting the language rules. So during the Christmas holidays in December 1989 (when his office was closed) he started a "hobby project" to create something better. He wanted a language that was simple, super readable (almost like English) and easy for anyone to pick up.

That hobby turned into Python and the first public version (0.9.0) came out in February 1991. Guido even named it after the British comedy show *Monty Python's Flying Circus* because he thought it sounded fun and unique nothing to do with snakes.

Python was built with these main goals in mind:-

- Really simple and clear syntax that anyone can understand quickly
- Code that's easy to read (indentation instead of braces helps a lot)
- Way less complicated than C++ or Java at the time
- You can do the same things with fewer lines of code

Because of all this Python made programming feel much more approachable. Over the years it became super popular thanks to its flexibility.

A.2 Licence

Python is released under the Python Software Foundation(PSF) License.It is an open source license which allows users to freely

use,modify and distribute the software.This means anyone can access Python's source code and make changes according to their needs.

The concepts of free software is based on four main freedoms:

- Freedom to run the program for any purpose
- Freedom to study how to program works
- Freedom to change it to fit your needs
- Freedom to share copies or your modified versiona

Python's license provides these freedoms making it flexible and widely used programming language.

A company can use Python modify it and even use it in their products but they must follow the license terms.Unlike some strict licenses like GPL the PSF license is more pmissive and does not require companies to share their modified code.

A.3 Ethics

For me open source is not just about licensing it's about the idea of sharing knowledge and helping others.It allows developers to learn from each other and improve software together and because of this it has become an important part of modern technology.

However not all software should be open source because on one side open source make learning easier,encourages collaboration and allows faster development.Anyone can access the code and improve it.But on the other side some software should remain closed source because of security and business reasons.Companies invest a lot of time,hardwork and money so they may be want to protect their work and maintain control over it.

Regarding the fact that whether it is ethical for a company to build a profitable product entirely on top of free community labor I think it can be both both ethical and questionable.It is ethical because open source licenses allow companies to use and modify the software but at

the same time it may be unfair if companies earn large profits without giving proper credit or contribution back to the community that developed the software.

The phrase “standing on the shoulders of giants” means building new ideas based on existing work.In software open source supports this idea because developers can use previous code and imporve it instead of starting from scratch.This helps innovation grow faster and makes development more efficient.However it also reduces originality as many projects depend on already existing work.

Overall open source plays a very important role in modern technology as it promotes sharing,learning and collaboration among developers.It makes software development more accessible and helps innovation grow faster.However it also raises important question about fairness,ownership and responsibility especially when companies use open source work for profit.Therefore while open source has many advantages it should be used in balanced and responsible way.

Part B — Linux Footprint

I performed this part of the project on my laptop using WSL (Windows Subsystem for Linux) since I use Windows 11. WSL provides a real Ubuntu environment inside Windows which allowed me to properly install and test everything as required. I opened Ubuntu from the Start menu updated the system and then performed all the steps.

Python is very common on Linux systems. Most distributions such as Ubuntu, Fedora and Mint already include Python 3 by default. It is widely used for scripting, automation and application development because it is simple and works well with Linux commands.

Installation Method
Python is usually pre-installed but it can also be installed using package managers:-

- On Ubuntu/Debian: sudo apt install python3
- On Fedora/Red Hat: sudo dnf install python3

I used sudo apt install python3 in my WSL Ubuntu and it installed quickly without needing any manual setup. Compiling from source is generally not required unless a specific version is needed.

Key Directories
After installation Python is located in standard Linux directories:

- /usr/bin/python3 → main executable
- /usr/lib/python3 → standard libraries
- /usr/lib/python3/dist-packages → additional installed packages

Python does not maintain large configuration or log files like services do. It works as a tool rather than a background process.

User and Group
Python runs under the current logged-in user. This improves security because it only has the permissions of that user and cannot modify system files without proper access.

Service Management

Python itself is not a service. It runs only when commands are executed. However, Python-based applications can be configured as services using systemd if needed. For this project, I only used Python as a command-line tool.

Update Model
Python updates are handled through the Linux package manager. I used:

- sudo apt update

- sudo apt upgrade

These commands install updates and security patches from Ubuntu repositories.

Overall Python works smoothly on Linux. It is lightweight, secure and easy to manage which makes it a very practical open-source tool.

Part C — The FOSS Ecosystem

Python does not work alone instead it is part of a large open-source ecosystem where many tools, libraries, and developers are connected. This is one of the main reasons why Python is widely used in areas like web development, data science, artificial intelligence, and automation.

What other open-source tools does it depend on?
Python depends on tools like the Cpython interpreter,pip for installing libraries and built in libraries such as math,os and sys.It also uses external libraries like Numpy,Pandas and Requests.

What Python has enabled or inspired
Python has helped develop many tools like Django and Flask for web development, TensorFlow and PyTorch for machine learning, and Jupyter Notebook for data work.

Relation to Web / LAMP Stack

Python is not part of the traditional LAMP stack, but it can be used instead of PHP. It can build web apps, connect to databases, and run on web servers

Community and Governance
Python has a large and active community:

- Developers discuss problems on forums and platforms
- Code is shared on GitHub
- Conferences like PyCon are organized

Python is managed by the Python Software Foundation (PSF) which ensures that it remains open source and freely available.

Overall Python has a strong ecosystem supported by tools, community and developers which makes it very useful and widely used.

Part D — Open Source vs Proprietry

Dimension	Open Source (Python)	Proprietary Alternative (MATLAB)
Cost	Completely free to download, use and share forever… No license needed at all. Great for students like me.	Paid software with licenses that can be really expensive. Hard for individuals or small groups.
Security (code audit)	Source code is open anyone can look inside, find bugs, fix them or check for security issues. Community helps make it safer over time.	Code is closed only MathWorks company can see and fix problems. We have to trust them completely.

Dimension	Open Source (Python)	Proprietary Alternative (MATLAB)
Support and reliability	Support comes from huge community: forums (like Stack Overflow), docs, YouTube tutorials, Reddit. It's reliable because so many people use and help with it but you might need to search yourself.	Official support from MathWorks with fast help, detailed guides and professional tools. Feels more reliable for big companies or urgent work.
Freedom to modify	I can change the code, add features or even make my own version if I want. This is why so many libraries (like NumPy, Pandas) exist.	I Can't touch the core code. MathWorks controls everything. No custom changes allowed.
Community vs corporate control	Run by the community and Python Software Foundation (PSF) decisions come from developers worldwide through proposals (PEPs). Everyone can contribute.	Fully controlled by the company (MathWorks). They decide updates, features and prices.

Dimension	Open Source (Python)	Proprietary Alternative (MATLAB)
Overall verdict	Python is flexible, free and super popular now (especially for AI, data science, and learning). It's the future for most people easy to start and grow with.	MATLAB is powerful for math/engineering tasks with nice built-in tools but cost and closed nature make it limited. Many are switching to Python.

CONCLUSION:

After comparing Python (open source) and MATLAB (proprietary) I feel Python is the better choice in most cases especially for students like me and for most developers.It's completely free, super flexible and backed by a huge active community. That makes learning programming easier, experimenting fun and building real projects possible without any money or restrictions. Plus Python has thousands of libraries and tools that cover almost everything: data science,web development, artificial intelligence,automation and more.

Of course proprietary software like MATLAB has some strong points too official customer support, very stable tools for specific engineering/math work and polished built in features. That can be helpful in professional or research settings where you need quick, reliable help from the company.

But overall I would definitely recommend Python for real world use. It's cost effective, highly versatile and lets anyone (students, hobbyists, startups, big companies) jump in and create without barriers.

In the future I really want to contribute to Python myself maybe by learning more deeply, fixing small bugs or building and sharing my

own little tools or scripts. Being part of the open-source community feels meaningful because you get to help others, learn from everyone and be part of something bigger than just one person or company.

3. Shell Script

Script 1 — System Identity Report (Unit 1 & 2)

```bash
#!/bin/bash

# Script 1: System Identity Report
# This script displays basic system information like a welcome screen
# --- Variables ---
STUDENT_NAME="Anushka"
SOFTWARE_CHOICE="Python"

# --- System Information ---
DISTRO=$(lsb_release -d 2>/dev/null | cut -f2)
KERNEL=$(uname -r)
USER_NAME=$(whoami)
HOME_DIR=$HOME
UPTIME=$(uptime -p)
DATE=$(date)

# --- Display Output ---
echo "======================================"
echo "   Open Source Audit - $STUDENT_NAME"
echo "======================================"
echo "Software : $SOFTWARE_CHOICE"
echo "Distro   : $DISTRO"
echo "Kernel   : $KERNEL"
echo "User     : $USER_NAME"
echo "Home Dir : $HOME_DIR"
echo "Uptime   : $UPTIME"
echo "Date     : $DATE"
echo "License  : GNU General Public License (GPL)"
echo "======================================"
```

```
========================================
   Open Source Audit - Anushka
========================================
Software : Python
Distro   : Ubuntu 24.04.4 LTS
Kernel   : 6.6.87.2-microsoft-standard-WSL2
User     : anushka
Home Dir : /home/anushka
Uptime   : up 20 minutes
Date     : Wed Mar 25 17:55:07 UTC 2026
License  : GNU General Public License (GPL)
========================================
```

This script displays basic system information such as kernel version,

user name, uptime and date. It uses variables, command substitution and echo statements. The purpose is to understand how system details can be accessed using shell scripting.

Script 2 — FOSS Package Inspector (Unit 2)

```
#!/bin/bash

# Script 2: Check if Python is installed

SOFTWARE="python3"

# Check installation
if dpkg -l | grep -q $SOFTWARE
then
    echo "$SOFTWARE is installed"

    # Get version
    VERSION=$($SOFTWARE --version)
    echo "Version: $VERSION"

    # Description using case
    case $SOFTWARE in
        python3)
            echo "Python: used for scripting, automation, and development"
            ;;
        *)
            echo "Unknown software"
            ;;
    esac

else
    echo "$SOFTWARE is not installed"
fi
```

```
python3 is installed
Version: Python 3.12.3
Python: used for scripting, automation, and development
```

This script checks whether Python is installed on the system.It used dpkg – l command along with grep to serach for the package.If the software is found it prints that it is installed and then displays its version using the –version command.

This script also uses a case statement to print a short description of Python explaining its purpose such as scripting,automation and development.If the software is not installed it displays a message saying that it is not installed.

This script demonstrates the use of conditional statements(if-else),case statements and basic package checking in linux.

Script 3 — Disk and Permission Auditor (Unit 2)

```bash
#!/bin/bash

# Script 3: Disk and Permission Auditor
# This script checks directory size and permissions

# List of directories
DIRS=("/etc" "/var/log" "/home" "/usr/bin" "/tmp")

echo "Directory Audit Report"
echo "----------------------"

# Loop through directories
for DIR in "${DIRS[@]}"
do
    if [ -d "$DIR" ]
    then
        PERM=$(ls -ld $DIR | awk '{print $1, $3, $4}')
        SIZE=$(du -sh $DIR 2>/dev/null | cut -f1)
        echo "$DIR => Permissions: $PERM | Size: $SIZE"
    else
        echo "$DIR does not exist on this system"
    fi
done

# Check Python config directory
CONFIG_DIR="/etc/python3"

if [ -d "$CONFIG_DIR" ]; then
    PERM=$(ls -ld $CONFIG_DIR | awk '{print $1, $3, $4}')
    echo "Config Dir ($CONFIG_DIR) => Permissions: $PERM"
else
    echo "Config directory not found"
fi
```

```
Directory Audit Report
----------------------
/etc => Permissions: drwxr-xr-x root root | Size: 4.3M
/var/log => Permissions: drwxrwxr-x root syslog | Size: 21M
/home => Permissions: drwxr-xr-x root root | Size: 52K
/usr/bin => Permissions: drwxr-xr-x root root | Size: 95M
/tmp => Permissions: drwxrwxrwt root root | Size: 24K
Config Dir (/etc/python3) => Permissions: drwxr-xr-x root root
```

This scripts checks multiple system directories such as /etc,/usr,/tmp, and /home.It uses a for loop to go through each directory and checks whether it exits.If the directory is present it displays its permission using the ls command and its size using the du command.If the directory does not exist it prints message accordingly.This scripts demonstrates the use of loop,conditional statements and basic file system commans in linux.

Script 4 — Log File Analyzer (Units 2 & 5)

```bash
#!/bin/bash

# Script 4: Log File Analyzer

LOGFILE=$1
KEYWORD=${2:-"error"}
COUNT=0

# Check if file exists
if [ ! -f "$LOGFILE" ]; then
    echo "Error: File $LOGFILE not found."
    exit 1
fi

# Retry if file is empty
while [ ! -s "$LOGFILE" ]
do
    echo "File is empty. Waiting..."
    sleep 2
done

# Read file line by line
while IFS= read -r LINE
do
    if echo "$LINE" | grep -iq "$KEYWORD"
    then
        COUNT=$((COUNT + 1))
    fi
done < "$LOGFILE"

echo "Keyword '$KEYWORD' found $COUNT times in $LOGFILE"

# Show last 5 matching lines
echo "Last 5 matching lines:"
grep -i "$KEYWORD" "$LOGFILE" | tail -n 5
```

```
Keyword 'error' found 3 times in /mnt/c/Users/Anushka/log.txt
Last 5 matching lines:
error: something failed
error: file not found
error: access denied
```

This script reads a log file and counts how many lines contain a specific keyword.It takes the file name and keyword as command line arguments and if no keyword is given it uses “error” by default.First it checks whether the file exists and if the file is not found it shows an error message.

The script then checks if the file is empty and waits until it has content.After that it read the file line by line using a while loop.For each line it used an if condition with grep to check if the keyword is present and increases a counter accordinglt.Finally it prints how many times the keyword appears and also shows the last five matching lines.This demonstrates the use of loops,conditions,file handling and command line arguments in shell scripting.

Script 5 — The Open Source Manifesto Generator (Unit 5)

```bash
#!/bin/bash

# Script 5: Open Source Manifesto Generator
# This script creates a small manifesto based on user input

echo "Create your own Open Source Manifesto"
echo "-------------------------------------"

# Taking input
read -p "Enter one open-source tool you use daily: " TOOL
read -p "What does freedom mean to you (one word)? " FREEDOM
read -p "What would you like to build and share? " BUILD

# Date and file name
TODAY=$(date)
FILE="my_manifesto.txt"

# Writing manifesto
echo "Date: $TODAY" > $FILE
echo "" >> $FILE
echo "I believe open source is about $FREEDOM." >> $FILE
echo "Tools like $TOOL help me learn and explore new things." >> $FILE
echo "In the future, I want to build $BUILD and share it with everyone." >> $FILE

# Output
echo ""
echo "Your manifesto has been saved in $FILE"
echo ""
cat $FILE
```

```
Create your own Open Source Manifesto
-------------------------------------
Enter one open-source tool you use daily: Python
What does freedom mean to you (one word)? Creativity
What would you like to build and share? a useful app

Your manifesto has been saved in my_manifesto.txt

Date: Wed Mar 25 18:42:21 UTC 2026

I believe open source is about Creativity.
Tools like Python help me learn and explore new things.
In the future, I want to build a useful app and share it with everyone.
```

This script creates a personalised open source manifesto using user input. It asks three questions and stores the responses in variables. The current date is taken using the date command. The script then combines all the inputs into sentences and writes them into a text file. Finally it displays the content of the file. This shows how user input, variables and file handling can be used together in shell scripting.

4. Reference Resources

1. GNU Project. *The Free Software Definition*
https://www.gnu.org/philosophy/free-sw.html

2. Open Source Initiative. *The Open Source Definition*
https://opensource.org/osd

3. Raymond, E. S. *The Cathedral and the Bazaar*
http://catb.org/~esr/writings/cathedral-bazaar/

4. Shotts, William. *The Linux Command Line*

https://linuxcommand.org

5. GNU Project. *Bash Reference Manual*

https://www.gnu.org/software/bash/manual/

6. SPDX License List

https://spdx.org/licenses/

7. Python Documentation

https://docs.python.org/3/